

Grade 6 Accelerated
Unit 15
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

For questions 1-4, use the word “**PROBABILITY**.”

1. What are the possible outcomes of choosing a letter randomly?
P, R, O, B, A, I, L, T, Y
2. Which outcomes have more than one possibility?
B, I
3. How many outcomes are included in the event that a consonant is chosen?
6 – P, R, B, L, T, Y
4. How many outcomes are included in the event that a vowel is chosen?
3 – O, A, I
5. The list of all possible outcomes for an experiment is called the

- ☐ event.
- ☒ **sample space.**
- ☐ set.
- ☐ probability.

For questions 6-8, use the twelve-sided dice shown.

6. What is the sample space?
1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12



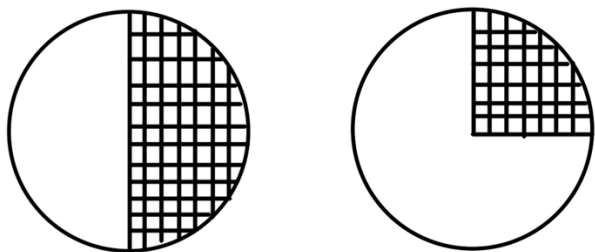
Sam and Steven are playing a game with the twelve-sided dice. For each game, determine and explain whether it is fair or not.

7. Sam wins if a prime number is rolled. Steven wins if a composite number is rolled.

This game is ☐ fair ☐ unfair because both outcomes have an equal chance of occurring.

8. Sam wins if a number 7 or less is rolled. Steven wins if a number 8 or greater is rolled. This game is ☐ fair ☐ unfair because Sam's outcome has a greater chance of occurring.

Use the two spinners shown below for questions 9-10.

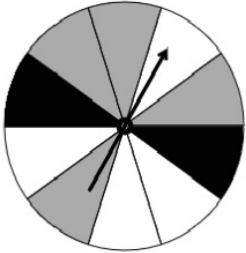


9. How many outcomes does each spinner have?
2

10. One of the spinners is fair and the other is unfair.

Spinner 1 is ☐ fair ☐ unfair because each section has an equal chance of being selected.

Spinner 2 is ☐ fair ☐ unfair because each section does not have an equal chance of being selected.

Simple Experiment	Game	Sample Space
Rolling a fair 6 sided number cube 	Player rolls a standard number cube and wins if it lands on a factor of 6.	$1, 2, 3, 4, 5, 6$
Spinning the spinner 	Player spins the spinner and wins if it lands on white.	$\text{Gray, black, white}$

1. What are the possible outcomes in the event of rolling a factor of 6 for Game 1?

$1, 2, 3, 6$

2. What fraction of the possible outcomes from Game 1 results in a win?

$$\frac{4}{6} = \frac{2}{3}$$

3. What are the possible outcomes in the event of spinning a white?

white

4. What fraction of the possible outcomes from Game 2 results in a win?

$$\frac{4}{10} = \frac{2}{5}$$

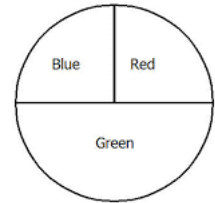
5. Which game does a player have a greater likelihood of winning?

Game 1

For each simple experiment, determine the sample space.

6. Spinning the spinner

Sample Space
red, blue, green



7. Prime numbers between 1 and 100

Sample Space
2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

8. Drawing an ace from a standard deck of 52 cards

Sample Space
Clubs – 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A repeat for spades, hearts, diamonds

9. Select a letter at random from the alphabet

Sample Space
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z

10. Selecting a random day of the week

Sample Space
Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday

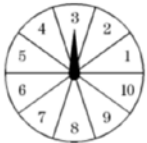
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Lesson 3 Practice

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1. Which of the following values could represent the probability of an event?

- A. $\frac{2}{5}$
B. 101%
C. -0.87
D. 6

Determine the probability of each event occurring as a fraction, decimal, and percent. Round decimals to the nearest hundredth if necessary.

	Event	Fraction	Decimal	Percent
2.	Landing on a prime number on a 12-sided die  P(prime)	2,3,5,7,11 are prime $\frac{5}{12}$	0.42	41.67%
3.	The spinner lands on a factor of 10  P(factor of 10)	1,2,5, and 10 are factors of 10. $\frac{4}{10} = \frac{2}{5}$	0.40	40%
4.	Pulling a yellow marble from a jar that has 5 blue, 4 green, and 8 red marbles. P(yellow)	5 + 4 + 8 = 17 total $\frac{0}{17}$	0	0%
5.	Picking a day of the week with the letter "t" in the spelling. P(t)	$\frac{3}{7}$	0.43	42.86%

Use the events from the table for questions 6-7.

6. Explain which event is least likely to occur.

Pulling a yellow marble from a jar with no yellow marbles is the least likely to occur at 0%.

7. Explain which event is most likely to occur.

Picking a day of the week with the letter "t" in the spelling is most likely to occur at 42.86%.

Samantha is practicing her multiplication facts using the flash cards below. If she chooses a card randomly, find the probability of each event as a fraction, decimal, and percent. Round decimals to the nearest hundredth and percentages rounded to the nearest tenth.

12×3	8×3	6×6	7×4	6×8
7×5	3×12	7×6	8×5	10×6
4×9	12×4	9×6	4×6	8×4

8. $P(\text{product of } 48) = \frac{2}{15} = 0.13 = 13.3\%$

9. $P(\text{product of less than } 40) = \frac{9}{15} = \frac{3}{5} = 0.60 = 60\%$

10. $P(\text{product not } 36) = \frac{11}{15} = 0.73 = 73.3\%$

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Lesson 4 Practice

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1. The probabilities of four events are shown. List them in order from least to greatest.

$$\frac{6}{13}$$

3 out of 9

41%

0.2

0.2,

3 out of 9,

41%,

$\frac{6}{13}$

Use the spinner to determine missing events, probabilities, and descriptions of likelihood. Express all probabilities as percentages. Use the vocabulary to complete each sentence.

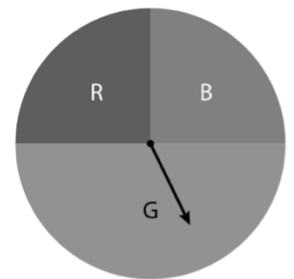
impossible

unlikely

equally likely as
unlikely

likely

certain



2. The probability of landing on green, $P(G)$, is 50%. This event is equally likely as unlikely.
3. The probability of landing on red, $P(R)$, is 25%. This event is unlikely.
4. The probability of landing on yellow, $P(Y)$, is 0%. This event is impossible.
5. The probability of landing on red or blue, $P(R \text{ or } B)$, is 50%. This event is equally likely as unlikely.
6. The probability of landing on a color, $P(\text{color})$ is 100%. This event is certain.

Here are some scenarios for questions 7-8.

- According to market research, a business has a 75% chance of making money in the first 3 years. 75%
- According to lab testing, $\frac{5}{6}$ of a certain kind of experimental light bulb will work after 3 years. $83\frac{1}{3}\%$
- According to experts, the likelihood of a car needing major repairs in the first 3 years is 0.7. 70%

7. Write the scenarios in order of likelihood from least to greatest after three years: the business makes money, the light bulb still works, and the car needs major repairs.

Car needs major repairs, business makes money, light bulb still works.

8. Name another chance experiment that has the same likelihood as one of the scenarios.

Answers will vary.

Ex: Using the spinner on the previous page, the likelihood of spinning and landing on green or red is 75%.

9. Which event is more likely: rolling a standard number cube and getting an even number, or flipping a coin and having it land heads up?

Since the probability of either event happening is 50%, both are equally likely.

10. Miami International airport reported that on a given day, the probability that an airline arrived on time was 0.32. The likelihood that the next airplane arrived earlier than scheduled was $\frac{3}{10}$ and the later than scheduled was 37.8%. What is most likely to occur when the next airplane arrives?

On time: 0.320

Early: 0.300

Late: 0.378

Because its probability is largest, it is most likely that the plane will arrive later than scheduled.

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Lesson 5 Practice

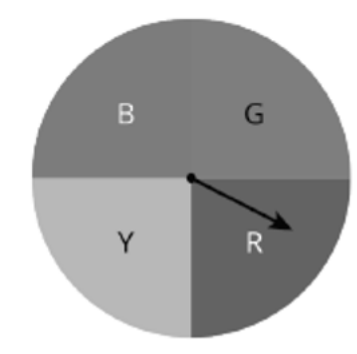
Name Answer Key
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1. Explain the difference between theoretical and experimental probability of a simple experiment.

Theoretical is $\frac{\text{outcome}}{\text{total events in the sample space}}$.

Experimental is $\frac{\text{times the event occurs}}{\text{total number of trials}}$.

Jennifer performed a simulation and spun the arrow on the spinner below. She recorded his results in the frequency table below.



Color	Frequency
Blue	8
Green	3
Red	7
Yellow	4

2. What is the theoretical probability of landing on green?

$$\frac{1}{4}$$

3. What is the experimental probability of landing on green?

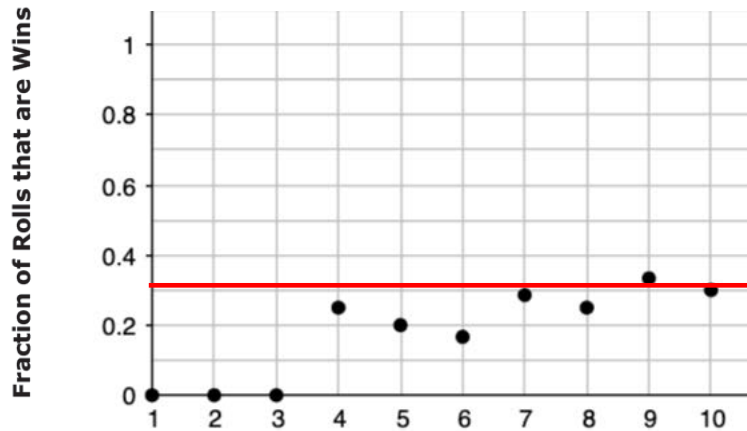
$$\frac{3}{8+3+7+4} = \frac{3}{22}$$

Tyler is playing a game where he wins if he rolls a 2 or a 5 on a standard number cube.

4. What is the theoretical probability that Tyler wins the game?

$$\frac{2}{6} \text{ or } \frac{1}{3}$$

He then performs the simulation 10 times. The graph below shows the fraction of games that are wins.



5. Estimate the probability of a roll winning this game based on the graph.

0.3

6. What type of probability is this?

Experimental

7. Draw a horizontal line on the graph representing the theoretical probability of rolling a 2 or a 5.

8. After 10 rolls, what fraction of the total rolls were a win?

$\frac{3}{10}$

9. If Tyler rolled the number cube 10 more times, estimate the probability that he will roll a 2 or a 5.

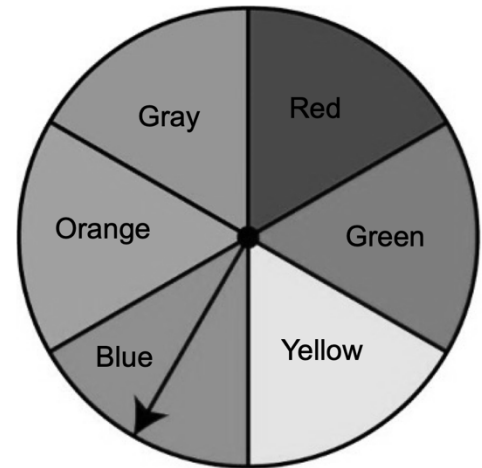
$\frac{1}{3}$

10. If he were to roll the number cube 500, how many times would you expect him to roll a 2 or a 5.

$\frac{1}{3}(500) \approx 167$ times

1. Find the theoretical probability of landing on each color as a percentage rounded to the nearest tenth.

Color	Theoretical Probability
Blue	$\frac{1}{6} = 16.7\%$
Yellow	16.7%
Green	16.7%
Red	16.7%
Gray	16.7%
Orange	16.7%



A computer simulator spinner performed more trials. The results as percentages are shown in the table below.

Color	Simulation 1 (10 trials)	Simulation 2 (100 trials)	Simulation 3 (1,000 trials)
Blue	10%	19.1%	16.7%
Yellow	30%	17.3%	16.7%
Green	10%	14.5%	16.7%
Red	20%	12.7%	16.7%
Gray	20%	13.6%	16.7%
Orange	10%	22.7%	16.7%

2. What do you notice in comparing Simulation 1 to Simulation 2?
With more simulations, the probabilities for each color get closer to the theoretical probability.
3. Complete the table by estimating the percentages for Simulation 3.
They should be closer to the theoretical probability the more times the spinner is spun.

Your teacher gives you a standard number cube with numbers 1–6 on its faces. You roll it 500 times. Your results are shown in the table below.

Outcome	1	2	3	4	5	6
Frequency	56	92	80	95	67	80

4. What is the experimental probability of rolling an odd number?

$$\frac{56+80+67}{500} = \frac{203}{500} = 0.406 \text{ or } 40.6\%$$

5. How many times would you expect to roll an odd number?

$$\frac{3}{6} = \frac{1}{2} \text{ or } 50\% \text{ of the time.}$$

6. As the number of trials in this simple experiment increases, the
gets closer to the

o experimental probability.
o theoretical probability.

o experimental probability
o theoretical probability

Four situations are described. For each situation, create a possible simulation to fairly represent it.

Answers may vary.

7. Suppose that, on average, a basketball player makes about three out of every four foul shots.
Flipping a coin where landing on heads represents making the foul shot, and tails represent missing it.
8. In three of the past five years, New York has had major snowstorms in March.
Flipping a coin could be used again. Heads means snowstorm in March and tails means no snowstorm. Flip the coin 5 times.
9. Your school is taking 6 buses of students on a field trip. Will you be assigned to the same bus that your science teacher is riding on?
Rolling a fair six-sided die to assign people to 1 of the 6 buses could be used.
10. An experiment will produce one of three different outcomes with equal probability for each. Why would using a coin to simulate the experiment be a bad choice?
Because a coin only has 2 outcomes but the experiment has 3. A fair six-sided die would be better because: rolling a 1 or 2 (first outcome); rolling a 3 or 4 (second outcome); rolling a 5 or 6 (third outcome).

The Jones family has 4 children and want one more child to complete their family. $\frac{3}{4}$ of the children are boys and the rest are girls.

1. What is the probability that the next child will be a boy, represented as a percent rounded to the nearest hundredth?

$$\frac{1}{2} = 0.5 = 50\%$$

2. What is the theoretical probability that the next child will be a boy?

$$\frac{1}{2}$$

Describe how each of the following can be used to simulate if the next child will be a boy.

3. A spinner with 8 equal sections

Answers may vary.

Mark half the sections for “boy” and half for “girl”

4. A twelve-sided die

Answers may vary.

Even numbers represent boy, odd numbers represent a girl.

5. A bag with 60 marbles

Answers may vary.

Choose $\frac{1}{2}$ to represent a boy and $\frac{1}{2}$ to represent a girl.

Spinning a spinner with 4 equal sized sections, one for each blood type.